

Integrated Action, Kinetic Bound, and the Weighted Metric Trace Lemma (Route δ , Plan 2, Building Block WEIGHTEDMETRICTRACE)

Abstract

This building block chains three ingredients into a pointwise metric distance bound at a radius $\hat{\rho}$ near $\rho_\delta = 1 - \kappa\sqrt{\delta}$, where throughout this note we write $\delta := \delta_T(\Omega)$. The repaired version does not use the unresolved centroid space-time Π estimate. Instead, on the levels where the isoperimetric defect is small it chooses a levelwise Fusco–Julin oscillation centre and uses the strong quantitative isoperimetric inequality to control both the Fraenkel distance and the normal oscillation. An oriented radial divergence lemma converts these into the homothetic L^1 trace required by [2]. The large-isoperimetric-defect levels are handled by Markov from the integrated deficit identity and the elementary uniform L^1 -Lipschitz bound (A0). This yields the integrated action bound $\int (d_{\mathcal{X}}(F_\rho, B_1)^2 + |\dot{F}_\rho|_{\mathcal{X}}^2) d\mu \leq C \delta_T$. A Talenti bad-set decomposition then gives the kinetic bound (K) in Lebesgue form, and the abstract weighted metric trace lemma produces $\hat{\rho} \in J^* := [\rho_\delta - L^*, \rho_\delta - L^*/2]$, where $L^* := (\rho_\delta - \rho_*)/4$, with $d_{\mathcal{X}}(F_{\hat{\rho}}, B_1)^2 \leq C_* \delta_T$. The chain uses no pointwise (R)/(1.2) and no centroid Π -closure input.

1 Statements

Fix $n \geq 2$, $R \geq 2$, $\rho_* \in (0, 1)$, $\kappa > 0$. Let $\Omega \subset B_R$ open with $|\Omega| = \omega_n$, $u = u_\Omega$ the variational torsion function, $\delta := \delta_T(\Omega) \leq \delta_0(n, R, \rho_*)$. Set $E_\rho := \{u > t(\rho)\}$ with $|E_\rho| = \omega_n \rho^n$, $\rho_\delta := 1 - \kappa\sqrt{\delta}$, $\mu(d\rho) := (-t'(\rho))d\rho$. Define $F_\rho := \rho^{-1}E_\rho \subset B_{R/\rho_*}$, $\gamma(\rho) := [F_\rho] \in \mathcal{X}$. For a.e. ρ , E_ρ has finite perimeter: this follows from the coarea formula applied to $u \in W_0^{1,2}(\Omega)$.

Theorem 1.1 (Integrated action bound).

$$\int_{\rho_*}^{\rho_\delta} (d_{\mathcal{X}}(F_\rho, B_1)^2 + |\dot{F}_\rho|_{\mathcal{X}}^2) d\mu(\rho) \leq C(n, R, \rho_*, \kappa) \delta_T(\Omega). \quad (1)$$

Theorem 1.2 (Kinetic bound (K), Lebesgue form).

$$\int_{\rho_*}^{\rho_\delta} |\dot{F}_\rho|_{\mathcal{X}}^2 d\rho \leq C'(n, R, \rho_*, \kappa) \delta_T(\Omega). \quad (2)$$

Theorem 1.3 (Weighted metric trace). *Set $L^* := (\rho_\delta - \rho_*)/4$ and $J^* := [\rho_\delta - L^*, \rho_\delta - L^*/2]$. There exist C_* and $\hat{\rho} \in J^*$ such that $d_{\mathcal{X}}(F_{\hat{\rho}}, B_1)^2 \leq C_* \delta_T(\Omega)$.*

2 Levelwise quantitative input

For a.e. ρ write

$$\mathcal{I}(\rho) := D_I(t(\rho)), \quad \mathcal{H}(\rho) := D_H(t(\rho)), \quad \Delta P_\rho := P(E_\rho) - P(B_\rho).$$

Since $P(B_\rho) \geq n\omega_n \rho_*^{n-1}$ and $D_I(t(\rho)) = (P(E_\rho) - P(B_\rho))(P(E_\rho) + P(B_\rho))$,

$$0 \leq \Delta P_\rho \leq C_{n, \rho_*} \mathcal{I}(\rho). \quad (3)$$

The exact level-set deficit identity of [1], with $dt = d\mu$, gives

$$\int_{\rho_*}^{\rho_\delta} (\mathcal{H}(\rho) + \mathcal{I}(\rho)) d\mu(\rho) \leq C(n, \rho_*) \delta_T(\Omega). \quad (4)$$

Lemma 2.1 (Fusco–Julin centre package). *There are constants $\eta_I > 0$, $C = C(n, R, \rho_*)$ and a bounded Borel centre field $\rho \mapsto z_\rho$ such that, on the good set*

$$G_I := \{\rho \in [\rho_*, \rho_\delta] : \mathcal{I}(\rho) \leq \eta_I\},$$

one has

$$|E_\rho \Delta B_\rho(z_\rho)|^2 \leq C \mathcal{I}(\rho), \quad (5)$$

$$\int_{\partial^* E_\rho} \left| \nu_\rho(x) - \frac{x - z_\rho}{|x - z_\rho|} \right|^2 d\mathcal{H}^{n-1}(x) \leq C \mathcal{I}(\rho). \quad (6)$$

Proof. Apply the strong quantitative isoperimetric inequality of Fusco–Julin [6] to E_ρ , whose volume is $|B_\rho|$. In scaled form it states that, for a suitable centre z ,

$$\left(\frac{|E_\rho \Delta B_\rho(z)|}{\rho^n} \right)^2 + \rho^{1-n} \beta(E_\rho, z)^2 \leq C_n \frac{\Delta P_\rho}{\rho^{n-1}},$$

where

$$\beta(E_\rho, z)^2 := P(E_\rho) - (n-1) \int_{E_\rho} |x - z|^{-1} dx.$$

For finite-perimeter sets the divergence identity

$$\int_{\partial^* E_\rho} \left| \nu_\rho - \frac{x - z}{|x - z|} \right|^2 d\mathcal{H}^{n-1} = 2\beta(E_\rho, z)^2$$

holds by testing $\operatorname{div}((x - z)/|x - z|) = (n-1)|x - z|^{-1}$ away from z and then truncating. Since $\rho \in [\rho_*, \rho_\delta]$, (3) gives (5) and (6). If η_I is fixed small enough, the Fraenkel term forces every such almost-minimizing centre to lie in a compact ball depending only on R and ρ_* ; outside G_I set $z_\rho = 0$. A standard measurable-selection argument ([9, §5.3]) gives a Borel choice. $\square$

Lemma 2.2 (Radial trace from oriented divergence). *For the centres of Lemma 2.1 and every $\rho \in G_I$,*

$$T_\rho := \int_{\partial^* E_\rho} \left| \frac{|x - z_\rho|}{\rho} - 1 \right| d\mathcal{H}^{n-1}(x) \leq C(n, R, \rho_*) \mathcal{I}(\rho)^{1/2}. \quad (7)$$

Proof. Translate so that $z_\rho = 0$ and write $e_r = x/|x|$. Set $w(r) := |r/\rho - 1|$ and $X(x) := w(|x|)e_r$. Since $E_\rho \subset B_R$ and z_ρ is bounded, $w(|x|) \leq C(R, \rho_*)$ on $\partial^* E_\rho$. Up to the irrelevant point $x = 0$ (when $z_\rho \in \overline{E_\rho} \cap \partial^* E_\rho$, the truncation argument below still applies since the singularity at $x = z_\rho$ is a single-point set, which is $\mathcal{H}^{n-1}$ -null in dimension $n \geq 2$),

$$1 = e_r \cdot \nu_\rho + (1 - e_r \cdot \nu_\rho),$$

therefore

$$T_\rho = \int_{\partial^* E_\rho} X \cdot \nu_\rho d\mathcal{H}^{n-1} + \int_{\partial^* E_\rho} w(|x|)(1 - e_r \cdot \nu_\rho) d\mathcal{H}^{n-1}.$$

The second term is controlled by the normal oscillation, since

$$1 - e_r \cdot \nu_\rho = \frac{1}{2} |e_r - \nu_\rho|^2.$$

For the first term use the finite-perimeter divergence theorem, first with the truncated smooth fields

$$X_{\varepsilon, M}(x) = \eta_\varepsilon(|x|) \chi_M(|x|) w(|x|) e_r$$

and then let $\varepsilon \downarrow 0$, $M \uparrow \infty$. The truncation is legitimate because X is bounded on B_R , its divergence is locally integrable for $n \geq 2$, and E_ρ has finite perimeter. Since $w(\rho) = 0$, the same boundary integral over B_ρ is zero, hence

$$\int_{\partial^* E_\rho} X \cdot \nu_\rho d\mathcal{H}^{n-1} = \int_{E_\rho \setminus B_\rho} \operatorname{div} X dx - \int_{B_\rho \setminus E_\rho} \operatorname{div} X dx.$$

For $r > \rho$,

$$\operatorname{div} X = \frac{n}{\rho} - \frac{n-1}{r} \leq \frac{n}{\rho},$$

while for $0 < r < \rho$,

$$\operatorname{div} X = \frac{n-1}{r} - \frac{n}{\rho}, \quad (-\operatorname{div} X)_+ \leq \frac{n}{\rho}.$$

Thus

$$\int_{\partial^* E_\rho} X \cdot \nu_\rho d\mathcal{H}^{n-1} \leq C(n, \rho_*) |E_\rho \Delta B_\rho|.$$

Combining the two estimates gives

$$T_\rho \leq C(n, R, \rho_*) \left(|E_\rho \Delta B_\rho(z_\rho)| + \int_{\partial^* E_\rho} \left| \nu_\rho - \frac{x - z_\rho}{|x - z_\rho|} \right|^2 d\mathcal{H}^{n-1} \right).$$

Finally use (5) and (6). Since $\mathcal{I}(\rho) \leq \eta_I \leq 1$ on G_I , this yields (7). $\square$

Lemma 2.3 (Good-level homothetic trace). *Let*

$$\bar{V}_\rho := \frac{1}{P(E_\rho)} \int_{\partial^* E_\rho} V_\rho d\mathcal{H}^{n-1} = \frac{P(B_\rho)}{P(E_\rho)}.$$

For every $\rho \in G_I$,

$$\left(\int_{\partial^* E_\rho} |H_{z_\rho, \rho} - \bar{V}_\rho| d\mathcal{H}^{n-1} \right)^2 \leq C(n, R, \rho_*) \mathcal{I}(\rho). \quad (8)$$

Proof. With $e_\rho(x) := (x - z_\rho)/|x - z_\rho|$,

$$|H_{z_\rho, \rho} - 1| \leq \left| \frac{|x - z_\rho|}{\rho} - 1 \right| + |e_\rho \cdot \nu_\rho - 1|.$$

Since $|e_\rho| = |\nu_\rho| = 1$, $|e_\rho \cdot \nu_\rho - 1| = \frac{1}{2}|e_\rho - \nu_\rho|^2$. Hence (6) and Lemma 2.2 imply $\int |H_{z_\rho, \rho} - 1| \leq C\mathcal{I}(\rho)^{1/2}$ on G_I . Moreover

$$P(E_\rho)|1 - \bar{V}_\rho| = P(E_\rho) - P(B_\rho) = \Delta P_\rho \leq C\mathcal{I}(\rho) \leq C\mathcal{I}(\rho)^{1/2}$$

since $\mathcal{I}(\rho) \leq \eta_I \leq 1$ on G_I . The triangle inequality gives (8). $\square$

3 The integrated action bound (5.2-int)

Proof of Theorem 1.1. Let $B_I := [\rho_*, \rho_\delta] \setminus G_I$. By (4),

$$\mu(B_I) \leq \eta_I^{-1} \int_{\rho_*}^{\rho_\delta} \mathcal{I}(\rho) d\mu(\rho) \leq C \delta_T(\Omega). \quad (9)$$

Distance term. On G_I , the quotient metric and the rescaling give

$$d_{\mathcal{X}}(F_{\rho}, B_1) \leq \rho^{-n} |E_{\rho} \Delta B_{\rho}(z_{\rho})|, \quad (10)$$

hence (5) implies $d_{\mathcal{X}}(F_{\rho}, B_1)^2 \leq C \mathcal{I}(\rho)$. On B_I we use only the trivial bound $d_{\mathcal{X}}(F_{\rho}, B_1) \leq 2\omega_n$, because both sets have volume ω_n . Therefore, by (4) and (9),

$$\int_{\rho_*}^{\rho_{\delta}} d_{\mathcal{X}}(F_{\rho}, B_1)^2 d\mu(\rho) \leq C \delta_T(\Omega). \quad (11)$$

Metric-derivative term. On G_I , Theorem T of [2], with the centre field of Lemma 2.1 and with the translation mode set to $a = 0$ (see [2, Thm. T] for the configurable form, in which the translation parameter a is exposed on the right-hand side of equation (T)), gives

$$|\dot{F}_{\rho}|_{\mathcal{X}}^2 \leq C \left[\left(\int_{\partial^* E_{\rho}} |V_{\rho} - \bar{V}_{\rho}| \right)^2 + \left(\int_{\partial^* E_{\rho}} |H_{z_{\rho}, \rho} - \bar{V}_{\rho}| \right)^2 \right].$$

The first square is the Cauchy–Schwarz defect controlled by [1]:

$$\left(\int_{\partial^* E_{\rho}} |V_{\rho} - \bar{V}_{\rho}| d\mathcal{H}^{n-1} \right)^2 \leq C(n, R, \rho_*) \mathcal{H}(\rho),$$

and the second square is (8). Thus

$$|\dot{F}_{\rho}|_{\mathcal{X}}^2 \leq C(\mathcal{H}(\rho) + \mathcal{I}(\rho)) \quad \text{for a.e. } \rho \in G_I.$$

On B_I we use the uniform metric Lipschitz bound (A0), $|\dot{F}_{\rho}|_{\mathcal{X}} \leq L_0(n, R, \rho_*)$. Combining this with (4) and (9) gives

$$\int_{\rho_*}^{\rho_{\delta}} |\dot{F}_{\rho}|_{\mathcal{X}}^2 d\mu(\rho) \leq C \delta_T(\Omega). \quad (12)$$

Equations (11) and (12) prove (1). $\square$

Remark 3.1. The good-level estimate is genuinely quantitative: the only square-root loss is the Fraenkel term from Fusco–Julin, and it is squared before integration. The large-defect levels are not estimated pointwise; their $d\mu$ -measure is paid for by the integrated isoperimetric defect.

4 The kinetic bound (K)

Lemma 4.1 (Uniform metric Lipschitz, (A0)). $|\dot{F}_{\rho}|_{\mathcal{X}} \leq L_0(n, R, \rho_*)$ a.e.

Proof. Nestedness gives $|E_{\rho'} \Delta E_{\rho}| \leq n\omega_n |\rho' - \rho|$. Split $F_{\rho} \Delta F_{\rho'}$ through $(\rho')^{-1} E_{\rho}$ and use the dilation estimate to get $d_{\mathcal{X}}([F_{\rho'}], [F_{\rho}]) \leq L_0 |\rho' - \rho|$. $\square$

Lemma 4.2 (Bad-set Lemma). $|B_{\tau_0}| := |\{\rho : -t'(\rho) < \tau_0 = \rho_*/(2n)\}| \leq K_2 \delta$.

Proof. On B_{τ_0} , with $\psi(\rho) = (-t'_B(\rho)) - (-t'(\rho)) \geq 0$, one has $|\psi(\rho)| \geq \rho_*/(2n)$. By the integrated Talenti gap $\int |\psi| d\rho \leq K_1 \delta$ (see [5] and the corollary in [1]), $(\rho_*/(2n)) |B_{\tau_0}| \leq K_1 \delta$. $\square$

Proof of Theorem 1.2. Split $[\rho_*, \rho_{\delta}] = G \sqcup B_{\tau_0}$ with G the good set where $-t' \geq \tau_0$. On G : $\int_G |\dot{F}_{\rho}|^2 d\rho \leq \tau_0^{-1} \int_G |\dot{F}_{\rho}|^2 (-t') d\rho \leq (2n/\rho_*) \cdot C\delta$. On B_{τ_0} : by (A0), $\int_{B_{\tau_0}} |\dot{F}_{\rho}|^2 d\rho \leq L_0^2 K_2 \delta$. Adding gives (2). $\square$

5 The weighted metric trace lemma

Theorem 5.1 (Abstract weighted metric trace). *Assume (H1) $\int (d_{\mathcal{X}}^2 + |\dot{\gamma}|^2) d\mu \leq C\delta$, (H2) $\mu([a, b]) \geq c(b - a) - C\delta$, (H3) $\mu_{\rho} \leq M$, (5.14) $\int |\dot{\gamma}|^2 d\rho \leq C'\delta$. Set $L^* := (\rho_{\delta} - \rho_*)/4$ and $J^* := [\rho_{\delta} - L^*, \rho_{\delta} - L^*/2]$. Then there is $\hat{\rho} \in J^*$ with $d_{\mathcal{X}}(\gamma(\hat{\rho}), B_1)^2 \leq C_*\delta$.*

Proof. Step 1: macroscopic Markov on a fixed sub-interval. By (H2), $\mu(J^*) \geq cL^*/4$. By (H1) + Markov on J^* , there is $\rho_0 \in J^*$ with $d_{\mathcal{X}}(\gamma(\rho_0), B_1)^2 \leq K_1\delta$ where $K_1 = 4C/(cL^*)$.

Step 2: kinetic transport on J^ .* For any $\hat{\rho} \in J^*$, $|\hat{\rho} - \rho_0| \leq L^*/2$, hence AGS Thm 1.1.2 + Cauchy–Schwarz give $d_{\mathcal{X}}(\gamma(\rho_0), \gamma(\hat{\rho}))^2 \leq |\hat{\rho} - \rho_0| \int |\dot{\gamma}|^2 d\rho \leq (L^*/2) \cdot C'\delta$ by (5.14).

Step 3: triangle. For every $\hat{\rho} \in J^*$, $d_{\mathcal{X}}(\gamma(\hat{\rho}), B_1)^2 \leq 2K_1\delta + L^*C'\delta =: C_*\delta$. $\square$

Verification of hypotheses for the torsion application:

- (H1) = Theorem 1.1.
- (H2) interval bound from Talenti / Nicola–Tilli profile gap [1].
- (H3) $-t' \leq 1/n$ via Talenti.
- (5.14) = Theorem 1.2.

Proof of Theorem 1.3. Apply Theorem 5.1 to $\gamma = [F_{\rho}]$. $\square$

Corollary 5.2 (Pointwise asymmetry at $\hat{\rho}$). *For the $\hat{\rho}$ selected in Theorem 1.3,*

$$|E_{\hat{\rho}}\Delta B_{\hat{\rho}}|^2 \leq C(n, \rho_*, R) \delta_T(\Omega).$$

Proof. By the rescaling $F_{\rho} = \rho^{-1}E_{\rho}$ recorded at equation (10), the metric distance bound $d_{\mathcal{X}}(F_{\hat{\rho}}, B_1)^2 \leq C\delta_T$ from Theorem 1.3 yields $|E_{\hat{\rho}}\Delta B_{\hat{\rho}}|^2 \leq C(n, \rho_*, R) \hat{\rho}^{2n} d_{\mathcal{X}}(F_{\hat{\rho}}, B_1)^2 \leq C(n, \rho_*, R) \delta_T(\Omega)$, which is the claimed pointwise asymmetry estimate. $\square$

6 Conditionality

- Theorem 1.1: consumes the exact level-set deficit identity of [1], the metric first-variation theorem of [2], the strong quantitative isoperimetric theorem of [6], and the oriented radial trace Lemma 2.2.
- Theorem 1.2: plus elementary (A0) and bad-set Lemma, both unconditional.
- Theorem 1.3: plus (H2), (H3) from Talenti.

Thus this block no longer depends on the centroid integrated Π -control input isolated in [3]. Pointwise (R)/(1.2) is never invoked.

References

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